

UNIVERSITY OF EDINBURGH  
COLLEGE OF SCIENCE AND ENGINEERING  
SCHOOL OF INFORMATICS

**INFR11145 TEXT TECHNOLOGIES FOR DATA SCIENCE PG  
AND INFR11229 UG**

**Monday 22<sup>nd</sup> May 2023**

**13:00 to 15:00**

**INSTRUCTIONS TO CANDIDATES**

- 1. Note that ALL QUESTIONS ARE COMPULSORY.**
- 2. DIFFERENT QUESTIONS MAY HAVE DIFFERENT NUMBERS OF TOTAL MARKS. Take note of this in allocating time to questions.**
- 3. CALCULATORS MAY BE USED IN THIS EXAMINATION.**

Year 4 Courses

Convener: K.Etessami

External Examiners: J.Bowles, A.Pocklington, R.Mullins

**THIS EXAMINATION WILL BE MARKED ANONYMOUSLY**

Please provide answers to **ALL** of the following questions. It is more important to have a clear answer than a longer answer. When calculations are needed, please **show steps** and give answer to **three** decimal digits.

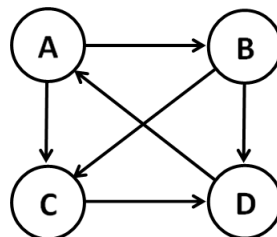
1. Provide a short answer to the following questions: [10 marks]

- (a) What are the components of a “test collection” in the Cranfield paradigm?
- (b) Why is smoothing important in language modelling?
- (c) What does the PageRank value, as discussed in the lecture, of a given web page represent?
- (d) Why don't commercial web search engines typically provide explicit relevance feedback functionality?
- (e) What is meant by a “shingle”?
- (f) What is meant by a “polite” crawler?
- (g) Can “eye tracking” be used as implicit feedback? If so, how?
- (h) What are the three major categories of user needs in Web Search?
- (i) How can an automatic search system detect that two queries are similar if they don't share a single word?
- (j) What is the role of the back queues in the crawler URL frontier?

2. Consider the following web pages and their out-links: [5 marks]

- Page A points to page B and C.
- Page B points to page C and D.
- Page C points to D.
- Page D points to page A.

Compute the PageRank scores at the end of the first two iterations for each of the pages. Teleportation factor is 0.5. Show your steps.



3. In the past few months, chatGPT and similar AI chat-bots have shown impressive performance. Some opinions claim that these chat bots will replace the web search engines. Do you agree with this opinion? Discuss in clear points why you agree or disagree. Justify your points with examples if possible. [5 marks]

4. **IR Evaluation** Search engine X produced the following ranked lists for queries A,B,C,D,E,F,G,H. Here a + means that a relevant document was found at the corresponding rank, while - means that a non-relevant document was found. Assume that all relevant documents were retrieved in the shown ranked lists for the 8 queries.

| rank: | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|-------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|
| A:    | + | - | - | - | - | - | - | - | - | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  |
| B:    | + | + | + | + | + | + | + | + | + | +  | +  | +  | +  | +  | +  | +  | +  | +  | -  | +  |
| C:    | + | - | + | - | - | - | - | - | - | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  |
| D:    | + | - | + | - | - | + | - | - | - | +  | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  |
| E:    | - | - | - | - | - | - | + | + | + | +  | -  | -  | +  | +  | +  | +  | -  | -  | -  | -  |
| F:    | - | - | - | - | - | - | - | - | - | -  | +  | +  | +  | +  | +  | +  | +  | +  | +  | +  |
| G:    | - | + | + | - | - | - | - | - | - | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  |
| H:    | - | - | + | + | - | - | - | - | - | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  |

- Compute the F1 score at rank 8 for query D. Report two decimal digits. [1 mark]
- Compute the R-precision for each of the 8 queries. [3 marks]
- Let nDCG' be a modified version of normalised Discounted Cumulative Gain (nDCG), which applies a linear discount to the gain instead of  $\log_2$  in the original formula, so that no discount is applied at rank 1, and discount keeps increasing linearly to contribute only 10% of the document's relevance at rank 10, then no contribution at all for documents ranked below 10..
  - Write down the formula for your definition of DCG' [2 marks]
  - Assuming "+" stands for gain = 1, and "-" stands for gain = 0. Calculate your nDCG' value at rank 10 for query D. [3 marks]
  - Explain what kind of search applications might find this formula more useful than the original ones. [1 mark]

5. You have collected a set of newspaper articles. Each article (document) belongs to exactly one category, such as *Germany* or *drink*. You have identified a number of words that are of interest to you, such as *London* and *match*, and counted how many documents in each category they appear in:

|                   | UK   | Germany | food | drink | politics | entertainment |
|-------------------|------|---------|------|-------|----------|---------------|
| Total no. of docs | 1024 | 1216    | 564  | 702   | 1020     | 605           |
| London            | 204  | 43      | 45   | 34    | 120      | 40            |
| Berlin            | 12   | 302     | 51   | 43    | 79       | 37            |
| Britain           | 561  | 69      | 56   | 45    | 204      | 120           |
| broccoli          | 43   | 45      | 122  | 90    | 5        | 3             |
| Indonesia         | 32   | 51      | 64   | 56    | 105      | 140           |
| voters            | 202  | 198     | 2    | 12    | 504      | 7             |
| match             | 67   | 67      | 12   | 12    | 89       | 321           |

Calculate the expected mutual information  $I(U; C)$  for the word *broccoli* and the category *food* (show steps).

$$\begin{aligned}
 I(U; C) &= \sum_{e_t \in \{1,0\}} \sum_{e_c \in \{1,0\}} P(U = e_t, C = e_c) \log_2 \frac{P(U = e_t, C = e_c)}{P(U = e_t)P(C = e_c)} \\
 &= \frac{N_{11}}{N} \log_2 \frac{N N_{11}}{N_{1.} N_{.1}} + \frac{N_{01}}{N} \log_2 \frac{N N_{01}}{N_{0.} N_{.1}} + \frac{N_{10}}{N} \log_2 \frac{N N_{10}}{N_{1.} N_{.0}} + \frac{N_{00}}{N} \log_2 \frac{N N_{00}}{N_{0.} N_{.0}}
 \end{aligned}$$

[5 marks]

6. Provide a short answer to the following questions:

[5 marks]

- What is the value of  $I(U; C)$  if term and class are completely independent?
  - What is the main limitation of probabilistic Latent Semantic Indexing (pLSI) compared with Latent Dirichlet Allocation (LDA)?
  - In Gibbs sampling for LDA, how do we assign a new topic to a word?
  - What is feature scaling?
  - In text classification, why do we need three separate datasets for training, development and testing?
7. In 2017, a news report appeared with the title “Google’s Sentiment Analyzer Thinks Being Gay Is Bad.” It describes a machine learning-based sentiment analysis system released by Google. The problem is that this tool was found to be biased. Sentences would receive different sentiment ratings based on which demographic characteristics or identity terms (such as “gay”) were mentioned in them, even when humans agreed that the sentiment in the sentences was the

*QUESTION CONTINUES ON NEXT PAGE*

*QUESTION CONTINUED FROM PREVIOUS PAGE*

same. For example, the system rated the sentence “I’m Christian” as slightly positive but the sentence “I’m a Sikh” as very positive, even though humans would rate both sentences as neutral.

[5 marks]

- (a) Explain, using your knowledge from TTDS, why you think this issue may have occurred.
  - (b) Come up with two alternative approaches that could be used to remedy this problem. For each of the two, discuss their advantages and disadvantages.
8. Your colleague is building a text classifier for detecting misinformation on social media. Her system will identify posts that will later be reviewed by human content moderators. She knows that her classifier won’t work perfectly, so she accepts if some posts that do not contain misinformation are identified. But it is very important for her that all posts that might contain misinformation are flagged by the algorithm, so they can be identified. She shows you some of her data:
- Misinformation: “Rio de Janeiro is the capital of Japan”
  - Not misinformation: “Tokyo is the capital of Japan”
  - Misinformation: “Joe Biden was born in Japan” “They are a really strong team who are doing really well”
  - Not misinformation: “Obama was born in Hawaii.”
- (a) How would you suggest she evaluate how well her classifier is doing? Recommend an evaluation metric and justify your choice.
  - (b) Give an example of a test sentence that her approach is likely to work on, and one that it is unlikely to work on. Justify your choice.

[5 marks]